

BIOL-8 Practice Test 12

Comprehensive Final Exam Preparation (Modules 01–17)

Instructions: This practice test samples five multiple-choice items per module (Modules 01–17) plus written items aligned with each module. Answer to the best of your ability.

Part A: Multiple Choice (85 questions)

Choose the best answer. Items 1–85 follow module order (01 through 17); Module 01 begins item 1.

Module 01: Exploring Life Science

1. Which of the following is a characteristic shared by ALL living things?

A) Cellular organization B) Sexual reproduction C) The ability to move from place to place D) The ability to photosynthesize
2. What is the correct order of biological organization, from smallest to largest?

A) Cell → Organ → Tissue → Organ System → Organism B) Cell → Tissue → Organ → Organ System → Organism C) Tissue → Cell → Organ → Organism → Organ System D) Organ → Tissue → Cell → Organ System → Organism
3. In a controlled experiment, the factor that the scientist deliberately changes is the:

A) Constant B) Dependent variable C) Independent variable D) Control group
4. What distinguishes a scientific theory from a hypothesis?

A) They mean the same thing in science B) Hypotheses are always proven correct before becoming theories C) Hypotheses are tested; theories are untested guesses D) A theory is supported by a large body of evidence from many experiments

5. Homeostasis refers to:

A) Maintaining a stable internal environment despite external changes B) The study of body structure C) Energy production in cells D) The process of cell division

Module 02: Chemistry of Life

6. The atomic number of an element is determined by its number of:

A) Neutrons B) Protons C) Protons + Neutrons D) Electrons

7. When atoms SHARE electrons, they form a:

A) Metallic bond B) Ionic bond C) Covalent bond D) Hydrogen bond

8. Water is considered a polar molecule because:

A) It contains hydrogen atoms B) It dissolves salt easily C) It has a high boiling point D) Electrons are shared unequally between oxygen and hydrogen

9. A solution with a pH of 3 would be classified as:

A) Strongly acidic B) A buffer C) Strongly basic D) Neutral

10. The four elements that make up approximately 96% of living matter are:

A) Au, Ag, Cu, Zn B) C, H, O, N C) Fe, Ca, K, Na D) He, Ne, Ar, Xe

Module 03: Biomolecules

11. The type of reaction that builds polymers by removing water is called:

A) Reduction B) Hydrolysis C) Dehydration synthesis D) Oxidation

12. Which macromolecule is the body's primary source of quick energy?

A) Nucleic acid B) Protein C) Lipid D) Carbohydrate

13. The monomers (building blocks) of proteins are:

A) Amino acids B) Nucleotides C) Fatty acids and glycerol D) Monosaccharides

14. What primarily determines the unique three-dimensional shape and function of a protein?

A) The temperature of the cell B) The sequence of amino acids C) The total number of carbon atoms D) The protein's location in the body

15. DNA and RNA are classified as:

A) Lipids B) Proteins C) Nucleic acids D) Carbohydrates

Module 04: Cellular Function

16. According to cell theory, which of the following is true?

A) All cells are the same size B) All cells have a membrane-bound nucleus C) All cells contain chloroplasts D) All cells come from pre-existing cells

17. The main difference between prokaryotic and eukaryotic cells is that:

A) Prokaryotes lack a membrane-bound nucleus B) Eukaryotes cannot reproduce C) Prokaryotes are larger than eukaryotes D) Eukaryotes lack ribosomes

18. Which organelle is the primary site of ATP production through cellular respiration?

A) Ribosome B) Mitochondria C) Golgi apparatus D) Nucleus

19. Ribosomes are the site of:

A) DNA replication B) ATP production C) Protein synthesis D) Lipid digestion

20. Which structure is found in plant cells but NOT in animal cells?

A) Plasma membrane B) Mitochondria C) Ribosomes D) Cell wall

Module 05: Membranes

21. The plasma membrane is described as a "fluid mosaic" because:

A) Proteins float and move within a flexible lipid bilayer B) It is rigid and tightly packed C) Water flows freely through it D) It is made entirely of phospholipids

22. The hydrophobic tails of phospholipids face:

A) Toward the cytoplasm B) Toward each other, inside the bilayer C) Toward the membrane proteins D) Toward the extracellular fluid

23. Cholesterol in the plasma membrane functions to:

A) Provide energy for active transport B) Transport oxygen across the membrane C) Maintain membrane fluidity across temperature changes D) Act as a receptor for hormones

24. Which type of membrane protein spans the entire lipid bilayer?

A) Glycoprotein B) Glycolipid C) Peripheral protein D) Integral (transmembrane) protein

25. A red blood cell placed in a hypotonic solution (like pure water) will:

A) Swell and possibly burst (lyse) B) Divide C) Shrink (crenate) D) Remain unchanged

Module 06: Metabolism

26. Metabolism is best defined as:

A) The breakdown of food in the stomach B) The sum of all chemical reactions in an organism C) The production of ATP only D) The release of heat from the body

27. A chemical reaction that releases energy (like breaking down glucose) is called:

A) Photosynthetic B) Endergonic C) Exergonic D) Anabolic

28. Enzymes speed up chemical reactions by:

A) Adding energy to the reaction B) Changing the end products C) Raising the temperature of the cell D) Lowering the activation energy required

29. The region on an enzyme where the substrate binds is called the:

A) Active site B) Catalyst site C) Product site D) Allosteric site

30. Why is ATP often called the cell's "energy currency"?

A) ATP permanently stores all of the cell's energy as fat B) ATP is identical to glucose C) ATP releases usable energy when its phosphate bonds are broken D) ATP is found only in red blood cells

Module 07: Genetics

31. Which molecule carries the genetic instructions for building an organism?

A) RNA B) Protein C) DNA D) Lipid

32. What are the four bases found in DNA?

A) Alanine, Threonine, Glycine, Cysteine B) Adenine, Thymine, Glucose, Cytosine C) Adenine, Uracil, Guanine, Cytosine D) Adenine, Thymine, Guanine, Cytosine

33. In DNA, Adenine (A) always pairs with:

A) Thymine (T) B) Uracil (U) C) Cytosine (C) D) Guanine (G)

34. What makes up the "backbone" of a DNA molecule?

A) Amino acids B) Sugar and phosphate groups C) Hydrogen bonds D) Nitrogenous bases

35. DNA replication is described as "semiconservative" because:

A) It requires no enzymes B) Only half the DNA is copied C) Each new DNA molecule contains one old strand and one new strand D) It happens in half the time of transcription

Module 08: Cell Division

36. During which part of the cell cycle is DNA copied?

A) G2 phase B) Mitosis C) G1 phase D) S phase

37. The correct order of the stages of mitosis is:

A) Prophase → Metaphase → Anaphase → Telophase B) Telophase → Anaphase → Metaphase → Prophase C) Metaphase → Prophase → Anaphase → Telophase D) Prophase → Anaphase → Metaphase → Telophase

38. During which stage of mitosis do chromosomes line up along the middle of the cell?

A) Prophase B) Metaphase C) Anaphase D) Telophase

39. At the end of mitosis and cytokinesis, the result is:

A) Two unique diploid cells B) Four identical haploid cells C) Two genetically identical diploid cells D) Four genetically unique haploid cells

40. What is the purpose of mitosis?

A) To increase genetic variation B) To reduce the chromosome number by half C) To produce sex cells (gametes) D) Growth, repair, and replacement of body cells

Module 09: Tissues and the Animal Body

41. The process by which the body maintains a stable internal environment is called:

A) Homeostasis B) Photosynthesis C) Fermentation D) Metabolism

42. A thermostat that turns on the heater when a room gets too cold is an example of:

A) Positive feedback B) Negative feedback C) Osmosis D) Diffusion

43. Maintaining the proper balance of water and salt in the body is called:

A) Metabolism B) Thermoregulation C) Osmoregulation D) Homeostasis only

44. Which organ system is most responsible for maintaining water and salt balance?

A) Muscular system B) Skeletal system C) Respiratory system D) Urinary (excretory) system

45. Breaking food into smaller pieces by chewing is an example of:

A) Mechanical digestion B) Absorption C) Fermentation D) Chemical digestion

Module 10: Inheritance

46. An alternate form of a gene is called a(n):

A) Chromosome B) Allele C) Genotype D) Phenotype

47. What is the difference between a genotype and a phenotype?

A) Genotype only refers to dominant traits B) They mean the same thing C) Genotype is the genetic makeup; phenotype is the physical appearance D) Genotype is the physical appearance; phenotype is the genetic makeup

48. An organism with two identical alleles for a trait (e.g., **BB**) is:

A) Hemizygous B) Polygenic C) Heterozygous D) Homozygous

49. An organism with two different alleles for a trait (e.g., **Bb**) is:

A) Heterozygous B) Codominant C) Homozygous dominant D) Homozygous recessive

50. Gregor Mendel is called the "Father of Genetics" because of his experiments with:

A) Bacteria B) Pea plants C) Mice D) Fruit flies

Module 11: Skeletal System

51. Which of the following is NOT a function of the skeletal system?

A) Support B) Protection of internal organs C) Hormone secretion for digestion D) Blood cell production

52. Where does hematopoiesis (blood cell formation) occur?

A) Periosteum B) Compact bone C) Yellow bone marrow D) Red bone marrow

53. Bones help regulate blood calcium levels. Which process releases calcium from bone into the blood?

- A) Bone resorption by osteoclasts B) Bone deposition by osteoblasts C) Cartilage formation
- D) Ossification

54. The shaft (long, central portion) of a long bone is called the:

- A) Epiphysis B) Diaphysis C) Periosteum D) Medullary cavity

55. The ends of a long bone are called the:

- A) Diaphysis B) Periosteum C) Epiphysis D) Endosteum

Module 12: Muscular System

56. Which of the following is a function of skeletal muscle?

- A) Lining the walls of blood vessels B) Controlling pupil size C) Pumping blood (involuntary)
- D) Moving the skeleton and generating body heat

57. Skeletal muscles help regulate body temperature because:

- A) Muscle contraction generates heat as a byproduct of ATP use B) They insulate organs from cold
- C) They absorb UV radiation D) They store large amounts of fat

58. The correct order from largest to smallest in muscle organization is:

- A) Myofibril → Fiber → Fascicle → Muscle B) Muscle → Fascicle → Fiber (cell) → Myofibril
- C) Fiber → Muscle → Fascicle → Myofibril D) Sarcomere → Muscle → Fascicle → Fiber

59. The basic contractile unit of a muscle fiber is the:

- A) Fascicle B) Myofibril C) Sarcomere D) Motor unit

60. The thick filaments in a sarcomere are made of:

- A) Troponin B) Tropomyosin C) Actin D) Myosin

Module 13: Nervous System

61. The central nervous system (CNS) consists of:

A) The brain and spinal cord B) The sensory organs only C) The autonomic nerves only D) All nerves outside the brain

62. The peripheral nervous system (PNS) includes:

A) The brain and spinal cord B) All nerves and ganglia outside the brain and spinal cord C) Only motor nerves D) Only the spinal cord

63. The division of the PNS that controls voluntary movements of skeletal muscles is the:

A) Autonomic nervous system B) Parasympathetic division C) Somatic nervous system D) Sympathetic division

64. The "fight-or-flight" response is controlled by the:

A) Central nervous system only B) Somatic nervous system C) Parasympathetic division D) Sympathetic division

65. Which part of a neuron receives incoming signals from other cells?

A) Dendrites B) Synaptic terminals C) Axon D) Cell body (soma)

Module 14: Microbiology

66. Prokaryotic cells differ from eukaryotic cells because prokaryotes:

A) Are always pathogenic B) Lack a membrane-bound nucleus C) Are larger than eukaryotic cells D) Have mitochondria

67. The genetic material of a typical bacterium is organized as:

A) RNA only B) Multiple linear chromosomes inside a nucleus C) A single circular chromosome in the nucleoid region D) DNA wrapped around histones in chromosomes

68. The three basic shapes of bacteria are:

A) Round, flat, and cubic B) Gram-positive, Gram-negative, and Gram-neutral C) Aerobic, anaerobic, and facultative D) Cocci, bacilli, and spirilla

69. Gram-positive bacteria differ from Gram-negative bacteria because Gram-positive bacteria have:

A) A thick peptidoglycan layer and no outer membrane B) No cell wall at all C) An outer membrane with lipopolysaccharide D) A thin peptidoglycan layer

70. Which statement about Gram staining is correct?

A) Gram-negative bacteria stain purple/violet B) Gram-positive bacteria retain the crystal violet stain and appear purple C) Gram staining identifies viruses D) Gram-positive bacteria stain pink/red

Module 15: Cardiopulmonary System

71. Oxygen-poor blood returning from the systemic circulation enters the heart first in the:

A) Right ventricle B) Left ventricle C) Right atrium D) Left atrium

72. From the right ventricle, blood is pumped toward the lungs through the:

A) Aorta B) Superior vena cava C) Pulmonary veins D) Pulmonary trunk (pulmonary arteries)

73. The first heart sound ("lub") is produced mainly when the valves between the atria and ventricles close. Those valves are the:

A) AV (atrioventricular) valves B) Valves in large leg veins C) Valves in the pulmonary veins D) Semilunar valves

74. The second heart sound ("dup") is associated most closely with the closing of the:

A) AV valves B) Semilunar valves C) Bicuspid valve only D) Tricuspid valve only

75. Which type of vessel carries blood **away from** the heart?

A) Veins B) Capillaries C) Arteries D) Lymphatic vessels

Module 16: Ecology

76. In most ecosystems, energy enters primarily as sunlight and is converted by:

A) Decomposers creating energy from heat alone B) Consumers synthesizing glucose without chlorophyll C) Herbivores fixing carbon from bone minerals D) Producers such as plants and algae

77. A food chain links producers to successive:

A) consumer trophic levels B) blood types C) abiotic reservoirs only D) photosynthetic pigments

78. Which pair best contrasts biotic versus abiotic factors?

A) Soil minerals versus altitude B) Trees versus rainfall C) Wind versus sunlight D) Temperature versus pH

79. Compared with energy, nutrients such as nitrogen often:

A) flow in one direction only without recycling B) increase in quantity at each trophic transfer C) cycle among organisms and the environment D) leave ecosystems permanently after one pass

80. Logistic population growth levels off mainly because environments have finite:

A) chromosome counts B) mutation rates C) oxygen only D) resources (carrying capacity)

Module 17: Evolution

81. Natural selection acts primarily on:

A) heritable variation affecting reproduction and survival B) random environmental noise unrelated to genes C) goals or intentions of organisms D) traits acquired during an individual adult lifetime

82. Which statement best distinguishes homology from analogy?

A) Homology applies only to bacteria B) Homology reflects shared ancestry; analogy reflects similar function by different ancestry C) Homology always means identical DNA sequences D) Analogous structures never look alike

83. Genetic drift has the strongest effects when populations are:

A) infinitely large B) heterozygous at every locus C) small D) free of mutation

84. Reproductive isolation tends to increase when:

A) gene flow is unlimited between groups B) two populations share identical mates every generation C) hybrids always have higher fitness than parents D) barriers reduce successful mating or viable/fertile offspring

85. Which fossil observation most directly supports common ancestry among vertebrates?

A) Transitional fossils showing intermediate body forms between major groups B) Species living at identical latitudes C) Same number of offspring per clutch D) Identical resting heart rates across all species

Part B: Fill in the Blank (34 questions)

Write the missing word(s). Two prompts per module (items 86–119).

Module 01

86. Science relies on evidence gathered through ___ and experimentation.

87. From molecule to biosphere, levels such as cell and ___ describe organization of life.

Module 02

88. Atoms bond by sharing electrons in ___ bonds.

89. Solutions with more H^+ than pure water are called ___.

Module 03

90. Enzymes are mostly ___ molecules that catalyze cellular reactions.

91. DNA stores genetic instructions using paired ___ bases.

Module 04

92. Plant cells often contain chloroplasts for photosynthesis; animal cells rely heavily on ___ for aerobic ATP.

93. The ___ directs traffic of proteins made on ribosomes.

Module 05

94. Passive transport moves substances ___ concentration gradients without ATP pumps.

95. Cells placed in hypotonic fluid tend to gain water by ___.

Module 06

96. Cells harvest chemical energy from glucose using pathways such as glycolysis and aerobic ___.

97. Coenzymes such as NAD^+ carry electrons during ___ breakdown.

Module 07

98. Transcription builds RNA using a DNA ___ strand.

99. Ribosomes decode ___ during translation.

Module 08

100. During ___, sister chromatids separate toward opposite poles.

101. Crossing over during ___ I exchanges segments between homologous chromosomes.

Module 09

102. ___ feedback loops oppose changes and help restore the body's set point (e.g., a thermostat-like response).

103. Most chemical digestion and nutrient absorption occurs in the ___ intestine.

Module 10

104. An organism with two different alleles is ___ at that locus.

105. Sex-linked traits involve genes carried on ___ chromosomes.

Module 11

106. Bone-removing cells called ___ degrade mineral matrix during remodeling.

107. Movement at many joints relies on lubricating ___ fluid.

Module 12

108. Contraction cycles require ATP, actin, ___, and calcium ions.

109. Each motor unit pairs one motor neuron with its muscle ___.

Module 13

110. Voltage-gated sodium channels help initiate action ___ along axons.

111. Neurotransmitters diffuse across the synaptic ___ after vesicle release.

Module 14

112. Gram-negative bacteria often have thinner peptidoglycan and an outer ___.

113. Horizontal gene transfer can spread antibiotic resistance via bacterial ___.

Module 15

114. The left ventricle pumps oxygen-rich blood into the ____.

115. Blood picks up most oxygen in pulmonary ____ surrounding alveoli.

Module 16

116. Food chains link producers to successive ____ levels.

117. Populations cannot grow forever because environments have finite ____.

Module 17

118. Natural selection acts on heritable ____ already present in populations.

119. Reproductive ____ separates populations on the path to new species.

Part C: Short Answer / Free Response (34 questions)

Answer clearly in complete sentences where appropriate (items 120–153). Two prompts per module.

120. Describe one way observation differs from opinion in biology lab work.

121. Explain why maintaining stable internal conditions (homeostasis) helps cells function.

122. Contrast ionic and covalent bonding using water or salt as an example.

123. Explain how pH relates to hydrogen ion concentration in everyday terms.

124. Relate protein shape to its ability to act as an enzyme.

125. Describe why nucleic acids store information better than simple sugars alone.

126. Compare roles of mitochondria and chloroplasts as energy-related organelles.

127. Explain why ribosomes exist in both prokaryotic and eukaryotic cells.

128. Differentiate passive transport from active transport with one example each.

129. Explain what happens to a red blood cell placed in very salty water.

130. Outline why cells need ATP even when resting.

131. Contrast aerobic respiration with fermentation in terms of oxygen use.

132. Trace information flow from DNA to functional protein in three short steps.

133. Define mutation and give one reason mutations matter for evolution.

134. Explain why mitosis preserves chromosome number while meiosis halves it.

135. Describe crossing over and state why it increases variation.

136. Give one example each of epithelial, connective, and muscular tissue roles.

137. Explain why cartilage differs from bone when resisting compression.

138. Solve conceptually: Why can carriers appear healthy for recessive disorders?

139. Explain pedigree symbols versus Punnett squares—what each tool shows best.

140. Describe how bones remodel after microdamage during exercise.

141. Explain synovial joint lubrication and why it reduces friction.

142. Explain role of calcium ions in starting muscle contraction.

143. Explain how motor unit recruitment lets a muscle generate more force as workload rises.

144. Explain resting potential versus depolarization at a neuron membrane.

145. Contrast sympathetic and parasympathetic effects on heart rate.

146. Explain why completing an antibiotic course reduces resistant survivors.

147. Differentiate bacteria from viruses regarding metabolism and replication.

148. Describe pulmonary circulation versus systemic circulation in one paragraph.

149. Describe gas exchange in the alveoli — name which gas moves into the blood and which gas moves out.

150. Explain energy flow versus nutrient cycling using a simple food chain.

151. Define carrying capacity and give one factor that lowers it.

152. Explain natural selection using variation, inheritance, and differential reproduction.

153. Describe one reproductive isolating mechanism that can lead to speciation.

End of Practice Test 12